

# KHALID SHAMS

## Enterprise Software Engineering Leader

Dhaka, Bangladesh (UTC+6) | Open to remote roles worldwide or on-site in Dhaka | One month notice  
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### PROFESSIONAL SUMMARY

Enterprise software engineer and solution architect with 16+ years across .NET, ERP, reporting, and SCADA-integrated platforms for utilities and industrial operations. Led delivery of systems serving 100k+ users and 1M+ monthly transactions with a 30+ engineer and QA team, cutting manual reporting effort by 65%. Now designs, builds, deploys, and operates three commercial systems in production single-handedly - a multi-tenant SaaS, an event-driven AI pipeline, and a storefront platform - covering architecture, containerized deployment, observability, and on-call.

### SELECTED PROJECTS

#### DentalPMS — Multi-tenant SaaS practice management for dental clinics

[demo.dentflowbd.com/login](https://demo.dentflowbd.com/login)

- Designed, built, deployed, and operate a commercial platform with paying clinics on it: 127k lines of C#, 41 domain entities, 167 automated tests, 33 EF Core migrations.
- Enforced tenant isolation structurally through EF Core global query filters at the DbContext, so a cross-tenant query cannot be written by accident rather than relying on developer discipline.
- Clean Architecture across four projects; Blazor Server UI over 81 components on PostgreSQL; JWT in HttpOnly cookies with policy-based authorization; nginx on a Linux VPS with per-clinic custom domains and 13 written runbooks.

ASP.NET Core 8 · Blazor Server · C# · PostgreSQL · EF Core · Docker · nginx · xUnit · Playwright

#### AI Job-Search Copilot — Event-driven AI matching pipeline with a polyglot microservice

[jobcopilot.dentflowbd.com/](https://jobcopilot.dentflowbd.com/)

[github.com/i-am-shams/ai-jobsearch-copilot](https://github.com/i-am-shams/ai-jobsearch-copilot)

- Built and operate a seven-container distributed system across two bounded contexts, with GitHub Actions building four images and deploying on push.
- Replaced a shared RabbitMQ queue with a fanout exchange after determining a second consumer would have silently round-robin away half of all live SignalR pushes; verified against per-queue message stats before and after.
- Added a Node.js/TypeScript notifications service on its own MongoDB Atlas cluster, adopted under Terraform by import rather than re-creation, with Grafana Alloy shipping metrics and logs to Grafana Cloud.

ASP.NET Core 8 · RabbitMQ · SignalR · React · TypeScript · Node.js · MongoDB Atlas · Terraform · Grafana Cloud · Docker Compose · GitHub Actions

#### One-Page Commerce — Single-product Cash-on-Delivery storefront platform

[one-page-commerce.vercel.app/](https://one-page-commerce.vercel.app/)

- Built and operate a commercial storefront platform with live stores selling through it, including an admin panel a non-developer can run unsupervised.
- Recompute every checkout price, discount, and delivery fee server-side from the database, so a client-posted total is never trusted.
- Next.js App Router with route-group separation and a tenant-scoped PostgreSQL schema, serving both single-tenant and pooled multi-tenant deployments from one codebase.

Next.js (App Router) · TypeScript · PostgreSQL · Tailwind CSS · Docker Compose

### PROFESSIONAL EXPERIENCE

#### Solution Delivery Specialist — Today's Tech

Apr 2025 - Present

- Own end-to-end solution architecture — from discovery and estimation through architecture alignment, implementation, quality gates, and production release
- Translate business goals into technical roadmaps, system designs, release plans, and measurable delivery outcomes
- Set technical direction across product, engineering, QA, and operations to keep enterprise delivery predictable and transparent
- Own sprint planning, backlog prioritization, dependency management, delivery risks, and release readiness
- Establish delivery governance through status reporting, stakeholder demos, risk tracking, and proactive mitigation

#### Technical Project Manager — BJIT LTD

Oct 2016 - Mar 2025

- Architected the CWASA Digital Ecosystem as project lead — billing, NRW analysis, management dashboards, predictive analytics, SCADA integration, GIS integration, and reporting workflows
- Delivered systems supporting 100k+ users, 1M+ monthly transactions, and a 65% reduction in manual reporting effort
- Conducted architecture reviews, code reviews, implementation guidance, dependency management, and production-readiness planning
- Designed and developed PDF creation software, educator content tools, and bridge inspection management systems

- Led a 30+ member cross-functional team of developers and QA engineers across planning, implementation, review, and release

#### **Software Engineer — Intelligent Image Management Inc.**

*Jun 2015 - Oct 2016*

- Implemented OCR and intelligent data-capture solutions using the Tesseract OCR engine
- Developed image-processing and extraction workflows for scanned documents and image-based inputs
- Improved structured data capture accuracy and reduced friction in document digitization workflows

#### **Software Engineer — Simplexhub Ltd**

*Aug 2013 - Jun 2015*

- Developed e-commerce platforms with integrated payment gateways and secure transaction workflows
- Created medical service management solutions for structured operational processes
- Implemented backend services, web interfaces, and data workflows for business-facing applications

#### **Software Engineer — LEADS Corporation LTD**

*Jul 2012 - Aug 2013*

- Developed financial analytics tools for stock-market-oriented data analysis
- Implemented real-time data-processing workflows for time-sensitive financial signals
- Created predictive logic and analytical features for market-trend decision support

#### **Lab Technical Officer — BRAC University**

*Jan 2012 - Jul 2012*

- Supported academic programming labs and workshops
- Maintained network infrastructure and server systems
- Provided technical guidance to students and faculty

#### **Programmer (Dot Net) — CACTS LTD**

*Apr 2010 - Dec 2011*

- Developed core ERP modules from scratch across inventory, HR, payroll, and administrative workflows
- Implemented relational database-backed business processes for enterprise operations
- Created reporting-oriented modules that improved internal workflow visibility and process control

### **TECHNICAL SKILLS**

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**Languages & Frameworks:** C# - ASP.NET Core - ASP.NET - Blazor Server - MVC - ADO.NET - WCF - Node.js - TypeScript - JavaScript/ES6 - HTML5 - CSS3

**Frontend:** React.js - Next.js - Angular - Blazor - Tailwind CSS - TanStack Query - Material-UI - Bootstrap - jQuery

**Databases & ORM:** PostgreSQL - MS SQL Server - Oracle - MySQL - MongoDB - Entity Framework Core - Entity Framework - Dapper - NHibernate

**Messaging & Real-Time:** RabbitMQ - SignalR - Event-Driven Architecture - Background Workers - WebSockets

**Architecture & Patterns:** Clean Architecture - Multi-Tenant SaaS - Microservices - RESTful API - Web API - SOAP - Repository - Factory - Singleton

**Testing & Quality:** xUnit - NUnit - MSTest - Playwright - Vitest - React Testing Library - Postman - Fiddler

**DevOps & Infrastructure:** Docker - Docker Compose - Terraform - GitHub Actions - Azure DevOps - Jenkins - nginx - Linux VPS Operations - Let's Encrypt / TLS - Git - TFS - SVN - NPM - Webpack

**Cloud & Services:** Azure Services - AWS Lambda - Serverless Architecture - Self-Hosted Production Deployment - MongoDB Atlas - Google Gemini API

**Observability:** Grafana Cloud - Grafana Alloy - Structured Logging - Liveness / Readiness Probes - Container Healthchecks

### **EDUCATION**

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**Bachelor of Science in Computer Science — BRAC University**

**Higher Secondary Certificate (HSC) — Rangpur Govt. College**

**Secondary School Certificate (SSC) — Cantonment Public School & College, Rangpur**

### **CERTIFICATIONS**

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Certified ScrumMaster® — SCRUM ALLIANCE® | Dec 2023 - Dec 2025 (not currently renewed)

### **LANGUAGES**

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Bengali (Native) | English (Fluent)